

Synopsis: Horizon Hysterical Response

Analytic agreement with Hawking radiation and Tolman temperature

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

Short guide to the horizon-response paper: HRT agrees analytically with Hawking and Tolman under the membrane/open-system embedding; Hawking and Tolman each touched that response from complementary sides.

1 Claim

Hysterical Response is the residual open-system response $F_H = -\text{Tr}(F\mathcal{L}^{-1}\mathcal{S})$ of an already-existing interaction. At a nonextremal horizon, with the exterior packaged as a dissipative membrane,

HRT horizon response + embedding and horizon regularity $\implies$ Hawking asymptotics and Tolman T_{loc} .
(1)

2 Hawking agreement

Literature Hypothesis 2.1 (Euclidean period). Regularity fixes $\beta_H = 2\pi/\kappa$.

Theorem 2.2 (Hawking temperature). $T_H = \kappa/(2\pi)$; *Schwarzschild gives $T_H = 1/(8\pi M)$.*

Theorem 2.3 (Hawking occupation). *Detailed balance at β_H yields $n_\omega = 1/(e^{\beta_H\omega} - 1)$ and the greybody-filtered asymptotic flux.*

Theorem 2.4 (Effective-method agreement with Hawking). *Under the membrane/open-system embedding and horizon KMS/FDR, the HRT calculation agrees analytically with Hawking's asymptotic thermal prediction: same T_H , Planck factor, and greybodies.*

3 Tolman agreement

Theorem 3.1 (Tolman local temperature). *For static observers at $r > 2M$ with $T_\infty = T_H$, $T_{\text{loc}}(r) = T_H/\sqrt{1 - 2M/r}$.*

Theorem 3.2 (Effective-method agreement with Tolman). *The same static-Killing scale continued by Tolman's law agrees with HRT's exterior thermal profile.*

No static Tolman temperature exists for $r < 2M$.

4 Why they touched HRT

Hawking computed the fluctuation branch at a causal dissipative boundary. Tolman computed the static equilibrium profile of that thermal scale. HRT joins both; the Euclidean period remains shared geometric input.

5 Evaporation

Theorem 5.1 (Integrated mass loss). $\dot{M} = -C/M^2$ integrates to $M^3 = M_0^3 - 3Ct$.